SANDRA
(Serve and Data Results Analysis)
Athletic Tech Glove

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Introduction

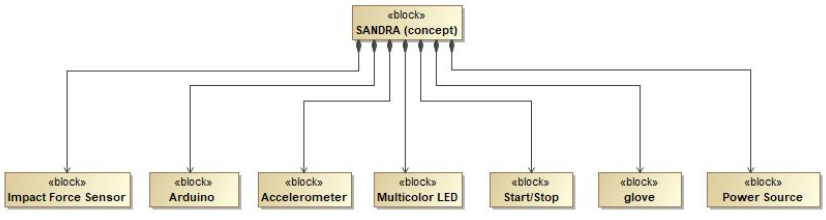
The goal of this project is to design and prototype a glove that collects telemetry in order to pinpoint areas of error in a volleyball serve. The idea of a glove came from the fact that today, there is nothing that measures physical performance that is self-contained, wearable, and easily available for all athletes to use outside of a lab environment. As many volleyball players struggle with their serve, and USA Volleyball's North Texas Region is home to nearly 130 competitive clubs[1], there is a need for an affordable, user friendly, accessible device that follows in the path of wearable fitness devices (such as FitBit).

The physics of a volleyball serve are not complex. At a basic level, a server must hit the ball with enough force and at a sufficient angle so that the ball clears the net both height- and distance-wise. The primary force that must be overcome during a serve is gravity. It is the force of impact that causes the ball to accelerate and leave the serving position at a high rate of speed. Collegiate volleyball players have reported an average jump serve speed of 19.7 m/s for males and 13.2 m/s for females [2].

The initial requirements were that the glove and its associated electronics (named SANDRA, for Serve and Data Results Analysis) should be wearable, lightweight, and have small, interchangeable components. Microcontrollers are compact integrated circuits designed to perform specific tasks. Microcontrollers typically combine a processor, read-only memory, and input-output functions on a single chip or a small printed circuit board [3]. Typical applications for microcontrollers include embedded systems that are part of manufacturing systems, robotics, automobiles, home appliances, office machines, medical and life support systems, and security systems, as well as internet of things (IoT) implementations.

Specifically, Arduino microcontrollers were selected as the basis for SANDRA. Arduinos are small, inexpensive, open-source, have a visual design platform (Visuino), and offer a broad selection of compatible sensors. While most microcontrollers are programmed utilizing the C coding language, the Arduino development environment is designed to hide most of the complexity from the programmer as it is designed for beginners. It only requires two calls to set up and run a continuous loop executing the intended task. This makes the Arduinos ideal for sensor data collection. An Arduino program is called a sketch, and is downloaded directly to the read-only memory of the Arduino, utilizing the Arduino IDE (integrated development environment). The Arduino and its compatible sensors laid the foundation for the conceptual design.

Figure 1: SANDRA Conceptual Block Definition Diagram



The conceptual design included an Arduino, an impact force sensor, an accelerometer, a switch to activate the glove, and an LED to provide visual feedback. The iterative process was then followed, wherein the design and requirements were refined as the constraints and capabilities of the components were discovered. After three iterations, the fourth and final design was reached.

Materials

Electronics	Assembly
1 Arduino Nano ($22.00)	2 Fingerless Gloves: 1 Left, 1 Right ($6.00)
15 ft long cable- USB A to Mini ($2.19)	1 Athletic Wrist Sweatband ($3.50)
1 MPU9250 Accelerometer ($6.80)	¾" wide Velcro ($0.80)
4 Female to Female Dupont Wires ($0.20)	3 adhesives ($0.25)
Software: Visuino, MagicDraw, Excel	Foam Core Board ($0.07)
PC	

Total Production Cost: $41.81

Procedure

1. Using MagicDraw, create a Block Definition Diagram and Internal Block Diagram of SANDRA to understand what pieces will be necessary for the construction, how they will connect, and what information will be sent in those connections.
2. Directly wire the electronics to design and test the needed circuitry and software to collect the data from the sensors.
3. Connect the accelerometer to the Arduino Nano in Visuino and check that the information it collects shows up properly in the designated section of the software.
4. Wire up the system prototype in Visuino and download the compiled Arduino sketch to the Arduino over the USB connection.
5. Confirm data is being collected utilizing the monitoring capabilities of Visuino.
6. Continue refining the circuitry and software, using an iterative process covering steps 1-5 to validate that the information being collected is accurate, consistent, and meaningful.
7. Calibrate the sensors to ensure data is within expected ranges.
8. Assemble the glove with the accelerometer mounted on the back of the glove and the Arduino mounted on the back of the wristband.
9. Validate that all data being collected remains accurate after the assembly.
10. Calibrate the accelerometer specifically for the serve by testing to see exactly what motion in the swing each axis measures.
11. Collect baseline characteristics for both successful and unsuccessful serves by having someone serve while wearing the glove.
12. Use the baseline data to determine rules for analyzing the serve data. Specifically, the rules should include a limit or range for the axial angles and g-force at impact.

13. Create an Excel spreadsheet using conditional formatting that highlights both failed axial and failed g-force results.
14. Configure conditional formatting such that a failure in the g-force measurement, as a serve's success is more dependent on g-force than on angle, or two axial failures will place a failed status for the serve in the spreadsheet.
15. Load the Arduino Sketch for data collection to SANDRA over the USB.
16. Collect live serve data that is displayed in the Visuino terminal console as formatted text.
17. Locate the sample with the highest g-force. Copy and paste the data for that sample into the spreadsheet for the analytics to occur.
18. Enter the visual success of the serve- whether or not the ball went over the net and inside the court- into the Visual Analysis column of the spreadsheet.
19. Repeat steps 16-18 for each serve.

Requirements

In the below table (Table 1), any requirement beginning with the number one (1) is a wearability requirement, while any requirement beginning with the number two (2) is a function requirement.

Number	Title	Description	Satisfied? Y/N
1	Wearability	SANDRA shall be wearable.	Y
1.1	Accelerometer	The accelerometer shall be mounted on the glove.	Y
1.2	Microcontroller	The microcontroller shall be mounted on a secondary article, worn above the wrist.	Y
1.3	Modularity	The electronics shall be interchangeable.	Y
1.4	Securely Mounted	The electronics shall be securely mounted.	Y
1.5	Securely Connected	The electronics shall be securely connected with Dupont wires.	Y
1.6	Soldered Pins	The soldered pins on the bottom of the electronics shall be covered in order to prevent short-circuiting and injury to the wearer.	Y
1.6.1	Pre-Soldered	All selected electronics shall be pre-soldered.	Y
1.7	Glove	The glove shall be fingerless, because of its athletic usage.	Y

1.7.1	Commercial	The glove shall be purchased from a sporting goods store.	Y
1.7.2	Ambidextrous	The glove shall be available in both right-handed and left-handed versions.	Y
1.7.3	Placement	The glove shall have no protruding pieces on the palm, since the palm is the place of impact.	Y
1.8	Power	SANDRA shall have a lightweight battery pack as a power source.	N
1.8.1	Battery Mounting	The power source (battery) shall be mounted on the secondary article.	N
1.9	Lightweight	SANDRA shall be lightweight in order to not inhibit the serve and its speed.	Y
1.9.1	Small Devices	The electronics shall be able to fit on the glove/secondary article properly.	Y
2.1	Acceleration	SANDRA shall be able to collect acceleration data of the hand while moving during the serving motion.	Y
2.2	Force	SANDRA shall be able to collect force data of the hand while moving during the serving motion.	Y
2.3	Data	SANDRA shall be able to transmit the collected data to a computing device in text format.	Y
2.3.1	Wireless	SANDRA shall be able to transmit data over bluetooth or wifi.	N
2.3.2	Server	SANDRA shall have an application that runs on the connected computer that listens for and collects the data transmitted from the Arduino.	Y
2.3.3	Analysis	Data analysis shall be performed in Microsoft Excel.	Y
2.3.3.1	Visual	The data analysis shall be able to provide a visual indicator of expected outcome based on comparison to the baseline value.	Y

2.4	Minimal Interface	The user shall be able to use SANDRA with little to no interaction with the device.	Y
2.5	Position Data	SANDRA shall be able to collect orientation of the hand through the serving motion.	Y
2.6	Accuracy	SANDRA shall be able to accurately report 90% or more of the serves.	Y
2.6.1	Precision	SANDRA shall collect the angular data with a precision of at least one degree.	Y
2.7	Battery Life	The battery shall be able to provide enough power for fifteen minutes of operation.	N

Table 1: SANDRA Function and Wearability Requirements

Data Results: Sensor Calibration

The sensor was calibrated by rotating it 90° along each axis, and voltage was captured at each stage. From this calibration, it was learned that 1g equates to 130mV. The calibration values were entered as ranges for mapped data values in the Visuino sketch.

Axis	0g Voltage	1g Voltage
X	270 mV	400 mV
Y	270 mV	400 mV
Z	270 mV	400 mV

Table 2: Sensor Calibration

Once the sensor was calibrated, a mechanism to convert the millivolts into degrees and aggregate G-force[4] was implemented, utilizing the sketch below in Figure 2.

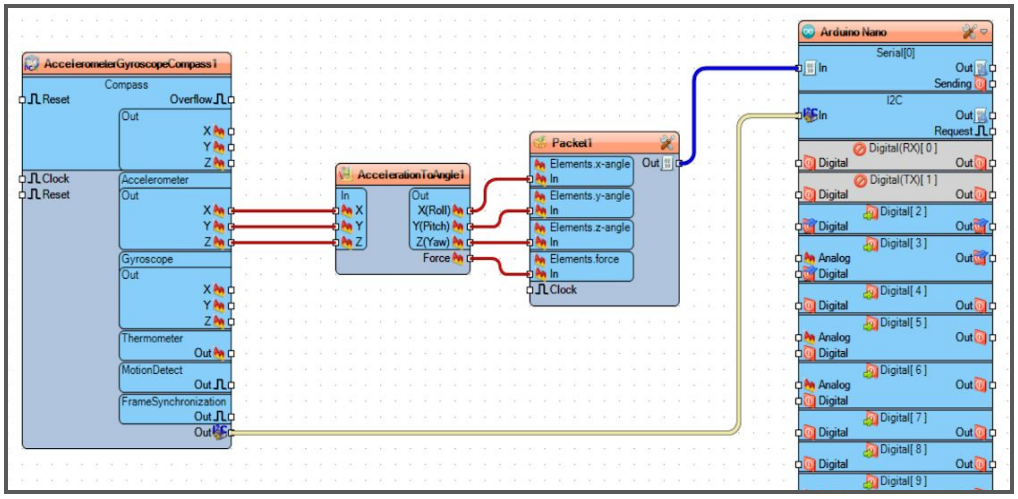


Fig.2: SANDRA Visuino Sketch for Serve Calibration

Figure 3 and Figure 4 display how the implementation of acceleration to angle function was validated, using the scope and instrumentation features that were programmed within the Visuino sketch.

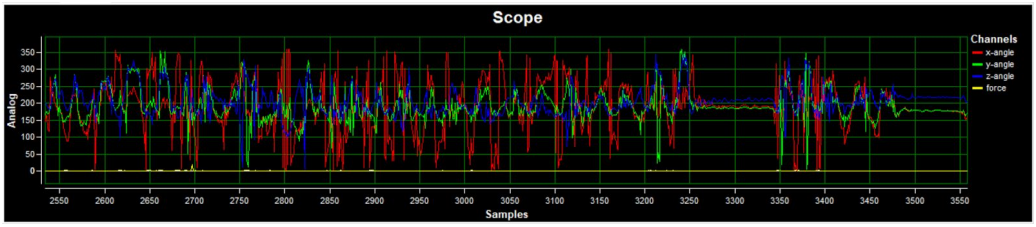


Fig. 3: Verification of Data Input from Sensor

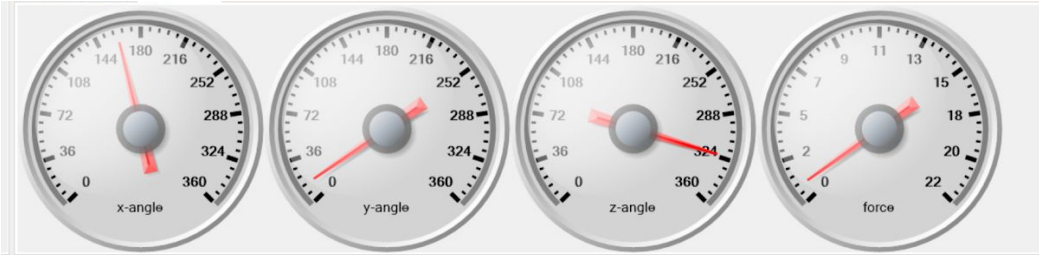


Fig.4: Post-Calibration Gauges, properly showing angular ranges from 0 to 360 degrees

Data Results: Serve Calibration

Using the data collection sketch shown in Figure 5, the serve was calibrated by having a server, wearing SANDRA, perform 15 serves- both successful and unsuccessful- in order to see the difference in the data.

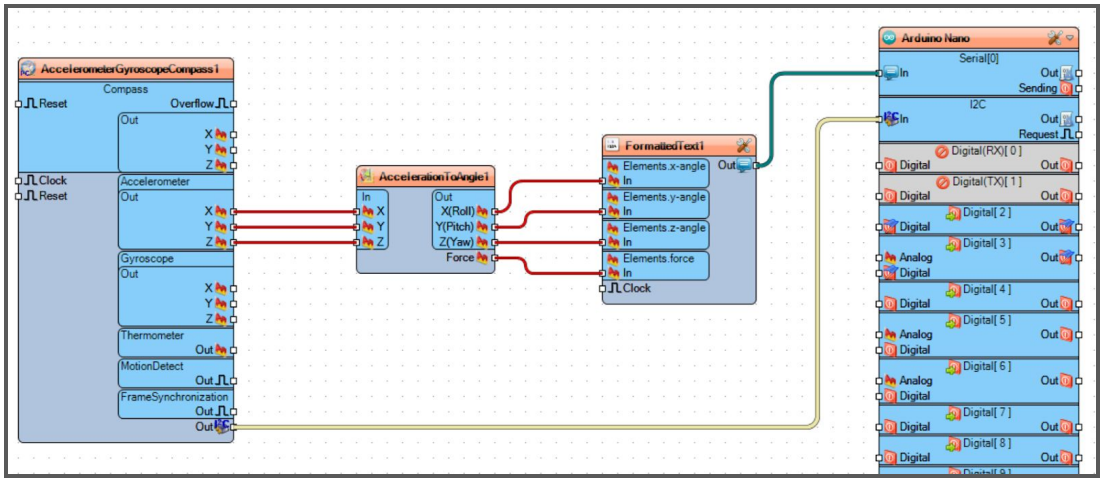


Fig. 5: SANDRA Visuino Sketch for Data Collection

Trial	X-Angle (degrees)	Y-Angle (degrees)	Z-Angle (degrees)	G-Force (g-s)	Success
1	257.85	215.73	188.80	18.48	1
2	302.08	315.00	32.08	24.75	0
3	225.00	290.67	290.67	23.42	1
4	266.49	224.48	183.45	19.50	1
5	263.49	222.41	185.95	18.96	0
6	45.00	135.00	135.00	27.71	0
7	302.18	213.51	157.38	20.20	1
8	265.58	221.02	183.85	21.24	1
9	269.25	220.62	180.65	20.95	1
10	265.44	225.00	184.56	22.66	1
11	302.05	221.26	151.23	23.03	0
12	264.71	221.75	184.73	20.00	0
13	262.32	228.16	188.57	19.90	0
14	267.56	218.85	181.96	20.30	1
15	315.00	225.00	135.00	27.71	1

Table 3: Serve Calibration Data

The table below (Table 4) displays the acceptable ranges of angles and g-force that constitute a successful serve. These rules were determined based on the data collected during serve calibration (Table 3).

Unit	X	Y	Z	G-Force
Measurement	Above 100º	Between 200º and 291º	Above 157º	Above 16g

Table 4: Rules for a Successful Serve

Data Results: Trials

An Excel spreadsheet with conditional formatting based on the rules shown in Table 4 was created to perform the collection and analysis of the data gathered via the electronics on the glove. The conditional formatting highlights failures and inserts an icon along the visual success, analysis, and accuracy columns. The collected data is presented below in Table 5.

Trials of 30 serves- 14 bad, 16 good- to test SANDRA's accuracy were conducted. The data analysis was validated by comparing it to the actual success of the serve, shown in the "Visual Success" column. SANDRA accurately reported the serve's status in 29/30 tests, yielding an accuracy of 96.7%. The only inaccurately analyzed serve was Serve #11 (refer to Table 5, below), which hit the tape on the top of the net. SANDRA might not be using a precision that can account for that scenario.

Trial	X-Angle (degrees)	Y-Angle (degrees)	Z-Angle (degrees)	G-Force (g-s)	Visual Success	Analysis	Accuracy
1	280.82	204.29	175.07	16.57	1	1	1
2	256.98	222.10	191.80	20.37	1	1	1
3	349.24	190.90	134.61	15.02	0	0	1
4	133.87	135.00	223.87	27.36	0	0	1
5	296.79	303.45	37.39	20.81	0	0	1
6	257.92	212.96	187.90	14.87	0	0	1
7	315.00	225.00	135.00	27.71	1	1	1
8	273.95	217.31	176.99	20.14	1	1	1
9	323.66	209.98	141.90	22.33	1	1	1
10	260.41	223.97	189.26	13.70	0	0	1
11	266.59	214.26	182.33	17.08	0	1	0
12	274.74	214.65	176.72	18.33	1	1	1
13	269.98	212.32	180.02	15.32	0	0	1
14	212.73	226.75	238.84	15.93	0	0	1
15	263.03	223.05	186.51	19.65	1	1	1
16	315.00	294.53	65.47	23.78	0	0	1
17	304.18	253.67	113.35	19.90	1	1	1
18	259.86	216.66	187.58	16.07	0	0	1
19	269.85	222.86	180.41	19.29	1	1	1
20	260.51	219.99	187.98	10.68	0	0	1
21	122.61	133.18	214.29	15.50	0	0	1
22	258.33	221.02	190.19	19.43	1	1	1
23	271.68	221.51	178.51	21.37	1	1	1
24	279.27	222.17	171.59	21.72	1	1	1
25	280.93	312.69	11.83	21.99	0	0	1
26	266.08	221.73	183.50	20.80	1	1	1
27	129.54	135.00	219.54	26.20	1	1	1
28	274.25	221.93	176.18	18.40	1	1	1
29	45.00	45.00	45.00	27.71	0	0	1
30	225.00	310.46	310.46	26.42	1	1	1

Table 5: Trial Results

Data Results: Requirements Coverage

Type of Requirement	Number of Requirements	Number of Satisfied Requirements	Number of Unmet Requirements	Percentage of Requirements Covered
Function	12	10	2	83.3%
Wearability	16	14	2	87.5%
Total	28	24	4	85.7%

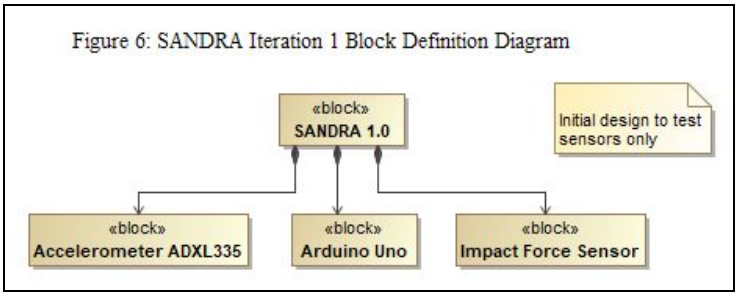
Table 6: Requirements Coverage

Discussion and Conclusions

The goal of this project is to design and prototype a glove that collects telemetry in order to pinpoint areas of error in a volleyball serve. The idea of a glove came from the fact that today, there is nothing that measures physical performance that is self-contained, wearable, and easily available for all athletes to use outside of a lab environment. Most tech that applies STEM to athletics requires a controlled environment, because the devices are external, usually requiring devices such as radar guns, photo-electric gates/sensor, or complex apparatus requiring significant computing power.

The initial requirements were that the glove and its associated electronics (named SANDRA, for Serve and Data Results Analysis) should be wearable, lightweight, and have small, interchangeable components. Due to these requirements, Arduino microcontrollers were selected as the basis for SANDRA instead of small PCs like a Raspberry Pi or a NUC. Arduinos are small, inexpensive, open-source, and offer a broad selection of compatible sensors. Additionally, the Arduino design platform, Visuino, is visual, which allows for rapid development. The Arduino and its compatible sensors laid the foundation for the conceptual design.

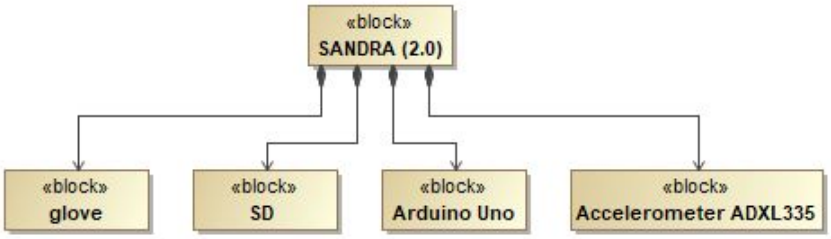
The conceptual design included an Arduino, an impact force sensor, an accelerometer, a switch to activate the glove, and an LED to provide visual feedback. The iterative process was then followed, wherein the design and requirements were refined as the constraints and capabilities of the components were discovered.



Iteration 1 (Figure 6), which consisted of an impact force sensor, Arduino Uno, and an accelerometer (ADXL335), was used only to test the sensors.

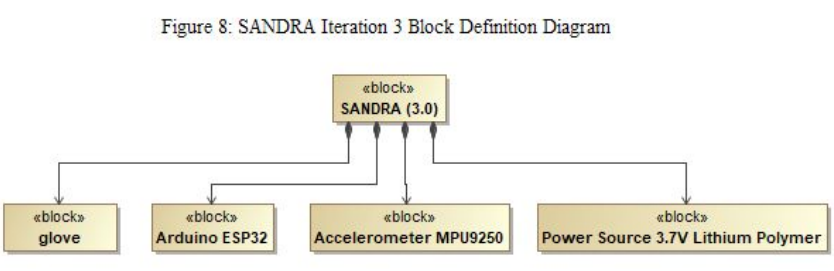
Subsequently, the impact force sensor was removed in Iteration 2 (Figure 7). The switch and the LED were also eliminated from the design in order to simplify the user experience as the Arduino did not have the necessary processing power to perform the analytics, making the visual indicator null. Acceleration-to-angle function was implemented in this stage. Iteration 2 consisted of a glove, an SD card, an ADXL335 accelerometer, and an Arduino Uno.

Figure 7: SANDRA Iteration 2 Block Definition Diagram



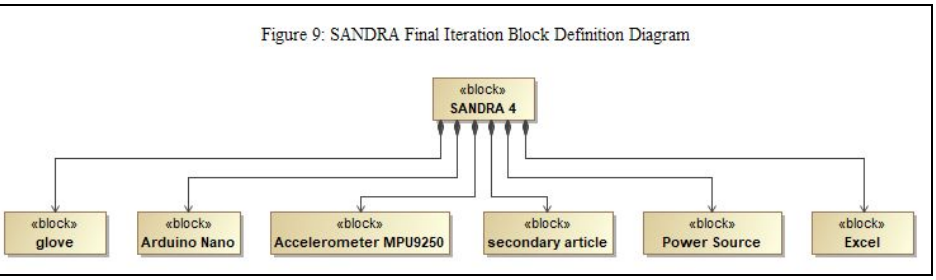
As wireless connectivity was desired to transmit data to a computing device, the Arudino UNO was replaced with an Arduino ESP32 in the third iteration (Figure 8). The ESP32 was chosen as it has both WiFi and low power Bluetooth capabilities. The accelerometer was changed to a sensor that utilized I^2C protocol, because the ESP32 does not have sufficient analog pins to collect data on all three axes from the ADXL335. The SD card for storage was eliminated because it was too large and required too much power.

Figure 8: SANDRA Iteration 3 Block Definition Diagram



After three iterations, the fourth and final design (Figure 9) was reached. In the final design, the Arduino Nano is connected to the MPU9250, which is an accelerometer and a gyroscope. The MPU9250 uses I^2C protocol, so it only requires one connection pin for data, and one for power.

Figure 9: SANDRA Final Iteration Block Definition Diagram



The accelerometer had to be calibrated and the data needed to be converted from millivolts to degrees, which is a form more meaningful for a SANDRA user. Therefore, the accelerometer was calibrated in order to implement acceleration-to-angle function, which translates the acceleration data from millivolts into angles and aggregate g-force.

The Arduino sketch used for sensor calibration was not ideal for serve calibration because it used a gauge, so two Visuino sketches were made: one for calibrating the sensor (Figure 2), and one for data collection (Figure 5). Using the data collection sketch, the serve was calibrated to create rules that define acceptable ranges of angles and forces for a successful serve. This was done by having a server, wearing SANDRA, perform 15 serves- both successful and unsuccessful- in order to see the difference in the data.

By testing different movements, it was discovered that the x-axis measures front/back movement of the hand (and the attached sensor), the y-axis measures left/right movement, and the z-axis measures up/down rotation of the wrist. The rules were then established: the x-angle must be above 100° to succeed, the y-angle must be between 200° and 291° to succeed, the z-angle must be above 157° to succeed, and the g-force must be above 16g to succeed. The calibration data showed that a serve's success is more dependent on g-force than on angle. A failed serve requires either an orientation failure along two axes or an insufficient g-force. The rules do not align one-to-one with the data that was collected during the controlled serve trials because it is impossible to perform every single possible combination of X, Y, and Z angles- roughly 5.8 million combinations - so the rules were derived from the data in conjunction with knowledge of a serve. The rules are believed to be within three degrees on each axis.

Development of the SANDRA prototype was successful, as 85.7% of the requirements were satisfied. The requirements were separated into two categories: wearability and function. Of the wearability requirements, which included general wearability, using all lightweight and small pieces, direct, interchangeable, and secure mounting of sensors, secure connections of wires/USB and electronics, covered pre-soldered pins to prevent short-circuiting and injury, commercial and ambidextrous availability of the fingerless glove, and having no protruding pieces on the palm, 87.5% were met. The unmet requirements were both related to having a lightweight battery pack as a power source. The lack of the battery did not impact the ability to create a successful prototype, because the USB cable provided power.

Ten of twelve function requirements, which included the ability to collect orientation on the X, Y, and Z axes, acceleration, and force data, transmitting it in textual format, rules for data analytics, and requirements for user interface, were satisfied, providing a total coverage of 83.3%. Since a battery was not implemented in the prototype, the requirement for 15 minutes of continuous operation from the battery was not achieved. Additionally, one requirement was only partially met because the data is being transmitted to a computing device, but not over bluetooth or wifi. Neither of these unmet requirements prevented completion of the prototype.

All requirements needed to adequately prototype SANDRA were completed. Most importantly, SANDRA's data collection and analytical requirements were both satisfied. Angular data provided sufficient precision to support the rule-set for an underhand serve. SANDRA correctly analyzed 96.7% of the 30 trial serves, exceeding the requirement for 90% accuracy.

Though SANDRA was successful, there are some possible sources of error. Error could have been introduced if the sensors themselves moved independently of the motion of the hand, thereby introducing noise. Another source of error could come from the fact that SANDRA operated under the idea that the point of impact coincides with the highest G-Force value

collected during the serve, but it is possible that is incorrect. The final source of error could come from the sensor's sampling rate of 20 samples per second, which means that the actual point of impact could be missed, though it is unlikely. This would impact the data collection and possibly the rules of a successful serve. The possible sources of error and the unmet requirements can be a basis for future work.

SANDRA was made specifically for situations outside of the lab, and virtually anywhere, due to its wearability. SANDRA can even be used somewhere without a net. There are five main aspects of future work that could be implemented. By completing the overall system with a battery pack and wireless capabilities, the unmet requirements in the original prototype will be satisfied. Sources of error such as noise can be dealt with by finalizing the electronics with actual wearable components to eliminate the need for interchangeability, and using wearable sensors embedded in the glove. Additions to the prototype can be to include mobile devices in a system that runs an app to which SANDRA transmits data for more sophisticated analytics to be performed. Also, by utilizing different sensors, different aspects of the serve can be measured, and all the data throughout the swing can be used to generate a visualization of the swing itself. SANDRA's capabilities will become more widespread by providing calibration for other types of serves, such as a standard overhand or a jump-serve, or for spot-serving. An official version of SANDRA could change the way volleyball players think about their serve and could greatly assist them, whether it is the first time or the thousandth time they have gotten a serve over the net. A volleyball game truly begins with a serve, and SANDRA can change the outcome of a game.

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Statement on Outside Assistance

Researcher's Name: Sophie Krajmalnik
Title of Paper: SANDRA (Serve and Data Results Analysis) Athletic Tech Glove
Where did you get the idea for your research? Many volleyball players- myself included- struggle with their serve, so I wanted to build a prototype to assist volleyball players with their serves.
Where did you conduct the major part of your work? I conducted most of my work at home.
If you worked in an institutional setting, did you work on your project as part of a team/group? I did not work in an institutional setting or as part of a team.
On what parts of your project did you receive help? I received help from Enrique Krajmalnik, the CTO of No Magic, Inc. He helped by narrowing down applicable online resources, manning the computer during all serve trials, and acting as a sounding board when problems were found. I also received help from Boian Mitov, the owner of Mitov Software and the developer of Visuino. He provided training on Visuino and excellent instructional videos. Both Enrique Krajmalnik and my mother assisted in preparing my written reports. Everything else I did on my own.
Is your research current or a continuation of previous research? This is not a continuation project.